

# VisionTrust Assurance

## Offline Computer-Vision Integrity & Provenance Framework

Report ID: REP-A1EABF1ED9C1 | Generated: 2026-09-23T15:22:24.024494+00:00 (Air-Gapped)

Overall Assurance Verdict

ACCEPT

Overall Risk Score: 0.0 / 100 (LOW)

Executive Summary: No integrity violations or anomalies detected.

| Metric                    | Value | Metric                | Value |
|---------------------------|-------|-----------------------|-------|
| Critical Findings         | 0     | Dataset Sample Count  | 49    |
| High Findings             | 0     | Dataset Classes       | 3     |
| Medium Findings           | 0     | Model Access Mode     | N/A   |
| Low Findings              | 0     | Model Parameter Count | N/A   |
| Inference Records Tracked | 7     | Tampered Inferences   | 0     |

## Integrity & Security Findings

No critical findings or integrity violations reported.

## Assurance Threat Coverage Statement

| ID  | Attack / Threat Capability            | Status              | Access Mode        | Primary Method                                   |
|-----|---------------------------------------|---------------------|--------------------|--------------------------------------------------|
| T1  | Label Flipping Detection              | SUPPORTED           | Dataset Labels + I | Feature k-NN neighborhood consensus & imbalan... |
| T2  | Systematic Mislabelling               | SUPPORTED           | Dataset Metadata + | Hierarchical contributor & batch anomaly roll... |
| T3  | Duplicate Flooding                    | SUPPORTED           | Image Files (Black | Perceptual Hashing (pHash, dHash) Hamming dis... |
| T4  | Out-of-Distribution (OOD) Injection   | SUPPORTED           | Dataset or Referen | Isolation Forest & statistical feature distri... |
| T5  | Trigger / Backdoor Injection          | PARTIALLY_SUPPORTED | Dataset Images     | Localized patch spatial consistency & class c... |
| T6  | Model Substitution                    | SUPPORTED           | Model File (.onnx, | Cryptographic SHA-256 identity & parameter co... |
| T7  | Model Weight Modification             | SUPPORTED           | Model Weights (Whi | Tensor statistics, layer shapes, and mean L2-... |
| T8  | Backdoor Behaviour Testing            | SUPPORTED           | Inference Executio | Controlled perturbation battery (brightness, ... |
| T9  | Inference Record Tampering            | SUPPORTED           | Full Record Payloa | Cryptographic hash binding + Ed25519 digital ... |
| T10 | Inference Replay Attacks              | SUPPORTED           | Inference Log Stat | Cryptographic nonces, monotonic sequence chec... |
| T11 | Provenance Configuration Substitution | SUPPORTED           | Provenance Metadat | Canonical JSON serialization hashing for prep... |
| T12 | Distribution Shift & Drift            | SUPPORTED           | Reference & Candid | PSI, 2-sample KS test, Wasserstein distance, ... |

## System Limitations & Air-Gap Compliance

- Framework runs 100% locally and does not transmit data or consult cloud foundation models.
- Statistical heuristics cannot mathematically prove total absence of steganographic or adversarial latent triggers.

- Model weight analysis on ONNX models depends on initializers embedded within the graph file.
- Offline cryptographic protection guarantees tamper-evident provenance but assumes private key is stored securely in air-gapped host.